

GOVERNMENT POLYTECHNIC

MUZAFFARPUR

DEPARTMENT OF

ELECTRONIC & COMMUNICATION ENGINEERING

A MAJOR PROJECT ON

"Smart Medication Management System"

SUBMITTED BY :-

1) *PURUSHOTAM KUMAR – 611172121602*
2) *SUDHANSHU KUMAR – 611172121603*
3) *RAJU KUMAR – 611172121029*
4) *SAINYURITA – 611172121605*
5) *PALLAVI KUMARI – 611172122044*
6) *KHUSHI KUMARI - 61117212260*

Smart Medication Management System

(using ESP32+IoT)

Project Report

CONTENTS

- INTRODUCTION
- OBJECTIVES
- KEY FEATURES
- COMPONENTS USED
- CIRCUIT DIAGRAM
- WORKING PRINCIPLE
- BLYNK INTEGRATION
- SAMPLE CODE SNIPPET
- ADVANTAGES
- LIMITATION
- USE CASESr
- FUTURE SCOPE
- CONCLUSION
- REFRENCES

1. Introduction

Medication adherence is a critical part of health care, especially for the elderly and chronically ill. This project, "Smart Medication Management System," aims to automate and improve the medicine reminder system using ESP32 and IoT integration via the Blynk platform. The system is designed to notify the user at correct times, dispense the required dose, and provide status updates on a mobile app. It also ensures functionality in the absence of Wi-Fi using physical controls

---

2. Objectives

- To automate daily medicine dispensing.
- To notify users of dose timing, missed doses, or taken doses via IoT.
- To allow manual and mobile app control of dispensing.
- To reduce dependency on caretakers or memory.
- To ensure the system works even when Wi-Fi is unavailable.

Key Features

- **Multi-Dose Scheduling (Up to 4 Doses Per Day)**

Users can configure four separate times each day. These are stored and monitored continuously, and each dose is triggered at the set time.

- **Blynk IoT Integration for Remote Control**

Users can control dose settings, check current status, manually dispense medicine, and receive alerts through a Blynk-based mobile app from anywhere.

- **Manual Control for Offline Mode**

The system provides offline capability through push buttons for all operations, ensuring reliability even in the absence of internet.

- **Wi-Fi/Offline Mode Switching**

A dedicated button lets users switch between online (Blynk) and offline (local control) modes. Feedback is given on the LCD and via RGB LED.

- **Real-Time Clock Integration**

The DS3231 RTC keeps accurate time independently of ESP32 power status.

- **Servo Motor-Based Dispensing**

Two servo motors rotate the dose carousel and open the gate to release medicine at the correct time.

- **IR Sensor to Detect Dose Pickup**

The IR sensor checks if the medicine was picked up within a specific time window after dispensing. Based on this, the system marks the dose as "Taken" or "Missed."

- **RGB LED and Buzzer Feedback**

Color-coded alerts:

- Red + Buzzer: Dose time.
- Green: Dose taken.
- Yellow: Offline mode.
- Blue: Connected to Wi-Fi.

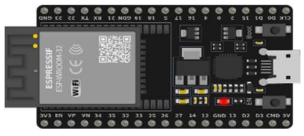
- **LCD Display Interface**

16x2 I2C LCD shows real-time info like system status, dose alerts, and connectivity state.

---

4. Components Used

Component	Quantity	Description
ESP32 Dev Board	1	32-bit MCU with built-in Wi-Fi & Bluetooth, ideal for IoT projects.
RTC DS3231	1	High precision Real-Time Clock with battery backup support.
I2C LCD 16x2	1	Displays system time, messages, and alarm status.
Servo Motor SG90	2	Controls medicine carousel rotation and gate opening mechanism.
IR Sensor Module	1	Detects motion or object (used to verify if medicine was taken).
Push Buttons	5	For manual time/dose adjustments dose selection and manual dispense.
RGB LED	1	Changes color based on system status (idle, active, missed dose).
Buzzer	1	Beeps on alarm triggers and warnings.
Power Supply	1	Ensures regulated 5V or 3.3V power for ESP32 and peripherals.
Jumper Wires	As required	For connections between modules and ESP32.



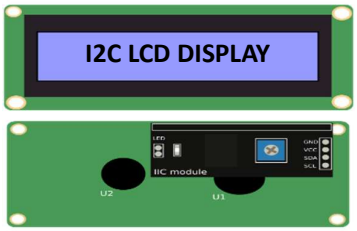
ESP32 MICROCONTROLLER

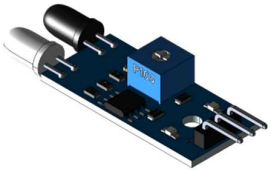


RTC MODULE

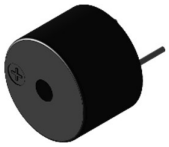


SERVO MOTOR

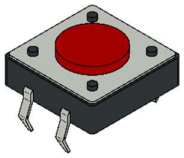




IR SENSOR



BUZZER



PUSH BUTTON



RGB LED



JUMPPER WIRES

5. Circuit Diagram

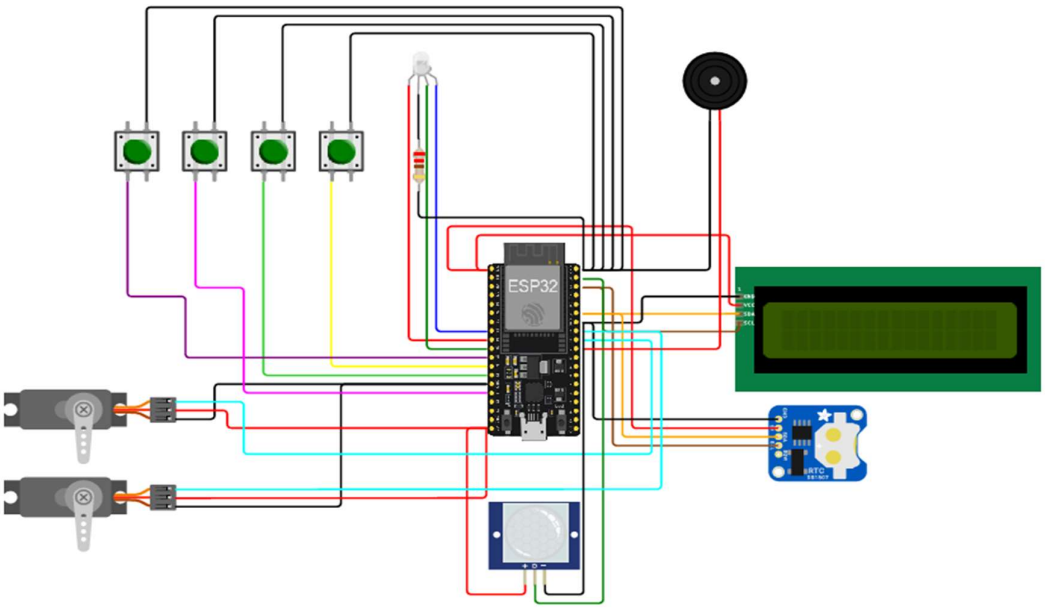


Fig: Circuit Diagram of Smart Medication System using ESP32

6. Working Principle

The Smart Medicine Dispenser is designed to ensure that patients take the right medicine at the right time—automatically and reliably, with or without internet connectivity. The system works in both **Wi-Fi (Online)** and **Offline** modes to ensure flexibility and uninterrupted functionality.

a) Initialization and Startup

- When powered on, the ESP32 microcontroller initializes all connected modules:
  - RTC (Real Time Clock) module for timekeeping
  - LCD for displaying time and system messages
  - Buzzer and RGB LED for alerts
  - Servo motors for dose dispensing
  - IR sensor for detecting if the medicine is picked up
  - Physical push buttons (for time setting, dose switching, and mode selection)
- The system attempts to connect to Wi-Fi and the Blynk IoT server. If successful, it enters **Wi-Fi Mode** by default.

b) Alarm Configuration

- Alarms can be set using either:
  - **Blynk App** (when in Wi-Fi Mode): The user selects the dose number (1–4) and sets the hour, minute, and AM/PM.
  - **Physical Buttons** (when in Offline Mode): Manual control buttons allow local time adjustments for each dose.
- All alarm times are stored in variables corresponding to dose 1–4.

---

c) Real-Time Monitoring

- The RTC module keeps track of the current time.
- The ESP32 continuously checks if the current time matches any of the configured dose times (for 1–4 doses).
- If a match is found, the corresponding **dose alarm is triggered**.

---

d) Dose Alarm Trigger

When a dose time is reached:

- **Buzzer sounds** and **RGB LED glows red** to alert the user.
- **LCD displays** a message like: "Time for Dose 2".
- If in **Wi-Fi Mode**, the same message is sent to the **Blynk App** using a virtual pin.

---

e) Dispensing Mechanism

- **Servo Motor 1** (carousel) rotates to align the medicine compartment for the current dose.
- **Servo Motor 2** (gate) opens to allow the medicine to be dispensed.
- A brief delay is added for the user to collect the medicine.
- The gate closes automatically after a set time.

---

f) Pickup Detection

- An **IR sensor** detects whether the medicine has been picked up from the tray.
- Two outcomes are possible:
  - ✅ **If picked up:**
    - LCD shows: "Dose Taken".
    - RGB LED turns green.
    - Blynk app displays a success message (if in Wi-Fi mode).
  - ❌ **If not picked up (within timeout):**
    - LCD shows: "Dose Missed".
    - RGB LED turns blue.
    - A missed dose notification is sent to the app (in Wi-Fi mode).

---

g) Offline Mode Functionality

- If **Wi-Fi is unavailable**, or the user **switches to Offline Mode** using the physical mode switch:
  - The system **bypasses Blynk functions**.
  - It continues to track time, trigger alarms, control the buzzer/LED, rotate servos, and detect dose pickup—all using offline logic.
  - Only the **manual buttons** are used for dose time adjustment and operation.
  - **LCD shows current mode:** "Offline Mode Active".

---

h) Wi-Fi Mode (When Available)

- Full control via the **Blynk mobile app**:
  - Set and update dose times.
  - View dose status (Taken/Missed).
  - Manual dose dispense via a button.
  - View current mode (Wi-Fi or Offline).
- The system also updates:
  - Current dose number
  - System status (On/Off)
  - Wi-Fi connection status

---

i) Manual Dose Dispensing

- A dedicated **manual dispense button** is provided both on the app and physically.
- When pressed:
  - The current dose is dispensed instantly.
  - Pickup detection logic still applies.
  - Status is shown on LCD and app (if connected).

---

j) . Smart Mode Switching

- A **physical push button** is used to switch between **Wi-Fi Mode** and **Offline Mode**.
- Default is Wi-Fi Mode.
- Pressing the button toggles the mode.
- The system remembers the selected mode during operation.
- LCD displays current mode status so the user is always informed.

---

k). Reset and Power Control

- A **System On/Off** control is available in the app (V6 virtual pin).
- When turned off, the device halts alarm checking and dispensing.
- Powering on resumes full operation.

🔄 Loop Behavior Summary

Step	Description
1	Read current time from RTC
2	Check selected mode (Wi-Fi or Offline)
3	Check if any dose alarm matches current time
4	If alarm triggered → buzz, light LED, show LCD/app message
5	Rotate servos to dispense medicine
6	Monitor IR sensor for pickup detection
7	Update status (Dose Taken/Missed)
8	Display result on LCD and Blynk (if online)
9	Wait for next loop cycle

---

7. Blynk Integration

Virtual Pins Mapping:

VPin	Function
V0	Alarm Hour
V1	Alarm Minute
V2	AM/PM Toggle
V3	Alarm Status Display
V4	Wi-Fi Status
V5	Manual Dispense
V6	System Power On/Off
V7	Current Dose Display
V8–V11	Dose Select (1–4)

---

9. Sample Code Snippet

```
if (alarmTriggered) {
    digitalWrite(buzzerPin, HIGH);
    setRGBColor("Red");
    lcd.setCursor(0, 0);
    lcd.print("Take Medicine" + String(currentDose));
    if (!isOfflineMode) {
        Blynk.virtualWrite(V3, "Take Medicine" +
String(currentDose));
    }
    dispenseDose(currentDose);
}
```

---

10. Advantages

- Ensures timely and reliable medication.
- Reduces human error or missed doses.
- Dual mode (Wi-Fi + Offline) for robustness.
- Easy to operate via mobile or manually.
- Can be extended for more doses or features.

---

11. Limitations

- Limited to 4 doses in current configuration.
- No backup power for ESP32 (can be added).
- Wi-Fi required for Blynk functions (though not mandatory due to Offline Mode).

---

12. Use Cases

- Elderly patients living alone
- Chronically ill patients with strict medication cycles
- Clinics and healthcare centers
- Rural or low-connectivity regions
- Hospitals with multiple patients

---

13. Future Scope

- Firebase cloud logging and syncing with multiple devices.
- Medicine stock level detection and refill alerts.
- Patient tracking, analytics, and caretaker integration.
- GSM module or FCM-based SMS notifications.

---

13. Conclusion

The Smart Medicine Dispenser is an innovative, reliable, and easy-to-use solution for managing daily medications. It merges IoT technology with a user-friendly interface to provide automatic reminders, dose dispensing, and real-time feedback. The newly added **Smart Mode Switching** ensures the system continues to serve its purpose regardless of network conditions.

---

14. References

- www.blynk.io (For IoT app-based control and virtual pin usage.)
- www.arduino.cc (For programming logic, libraries, and microcontroller integration.)
- https://wokwi.com/ (For simulating and testing the project virtually before hardware setup.
- ESP32 Datasheet (To understand ESP32's capabilities, pin configuration, and hardware specs.)